

- May cause call drop

✓ FINAL TIP (For Exam)

- Always draw hexagonal cell diagram
- Write 8 differences FDMA/TDMA/CDMA
- GSM diagram is very important (5–8 marks)
- Handoff + propagation diagrams = scoring

If you want, I can next:

- ✓ Convert this into PDF notes
- ✓ Give short 4-mark answers
- ✓ Or create last-day revision sheet 🚀

Sources



give in more detailed



Got it Rohit 👍 — here are **HIGHLY DETAILED, EXAM-READY NOTES (UNIT 3)** with definitions, diagrams, explanations, examples, advantages, disadvantages + PPT page references. You can directly write this in exams for 5–10 marks answers.

📘 UNIT 3: DETAILED NOTES (WITH PPT REFERENCES)

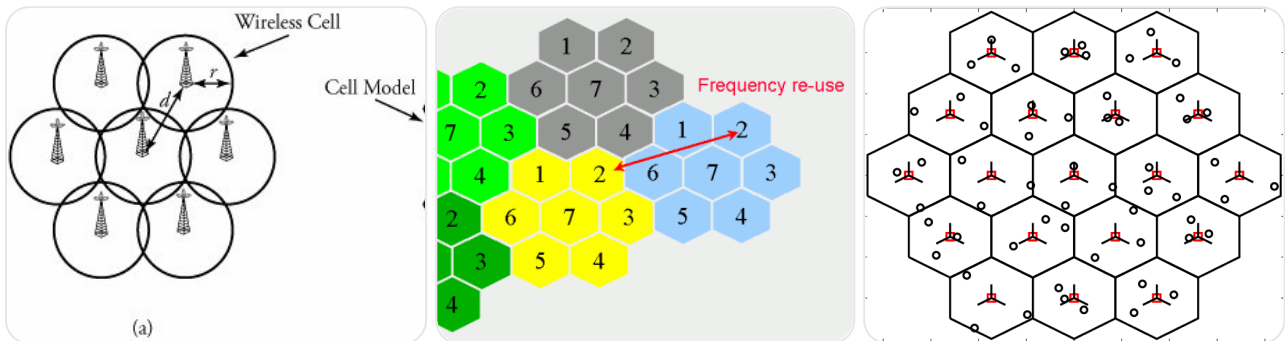
◆ 1. Cellular Concept in Mobile Communication

📄 PPT Pages: Cellular Concept section 📄 UNIT 3 (1)

✓ Definition

Cellular concept is a wireless communication technique in which a **large geographic area** is **divided into smaller regions called cells**, each served by a **Base Station (BS)** to provide communication services.

Diagram (Cells + Cluster)



Explanation

- Entire area is divided into **small hexagonal cells**
 - Each cell has a **base station (tower)**
 - Cells are grouped into **clusters**
 - Each cluster uses a **set of frequencies**
 - Same frequency reused in distant cells
-

Why Hexagonal Shape?

- Covers area without gaps
 - Simple mathematical design
 - Closest to circular coverage
-

Types of Cells (Hierarchy)

- **Macrocell** → Large coverage (city)
 - **Microcell** → Urban areas
 - **Picocell** → Buildings
 - **Femtocell** → Home
-

Cell Splitting

When users increase →

- Large cell → divided into smaller cells
 - Increases capacity
-

Real-Life Example

In Mumbai, telecom companies divide the city into **multiple cells** to support lakhs of users.

Advantages

- Efficient use of spectrum
 - High capacity
 - Low power usage
 - Supports mobility
-

Disadvantages

- Complex network planning
 - Requires handoff
-
-

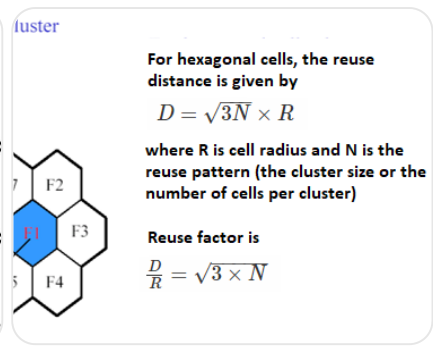
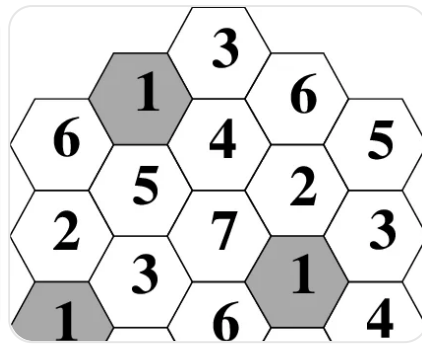
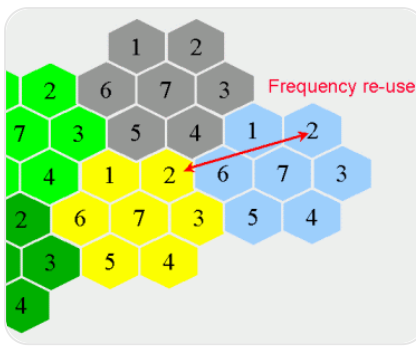
2. Frequency Reuse Concept

 PPT Pages: Cellular Concept / Frequency reuse  UNIT 3 (1)

Definition

Frequency reuse is the technique of **using the same frequency channels in different cells separated by a safe distance** to increase capacity.

Diagram



📌 Explanation

- Frequencies are limited resource
- Same frequency reused in **non-adjacent cells**
- Reduces need for new spectrum

📌 Important Terms

- **Cluster Size (N)** → Number of cells in cluster
- **Co-channel cells** → Cells using same frequency
- **Reuse distance (D)** → Distance between same frequency cells

📌 Formula

Reuse Ratio = D / R

🌍 Example

Same frequency used in **two distant towers in different parts of city**.

✅ Advantages

- Increases capacity
- Better spectrum utilization

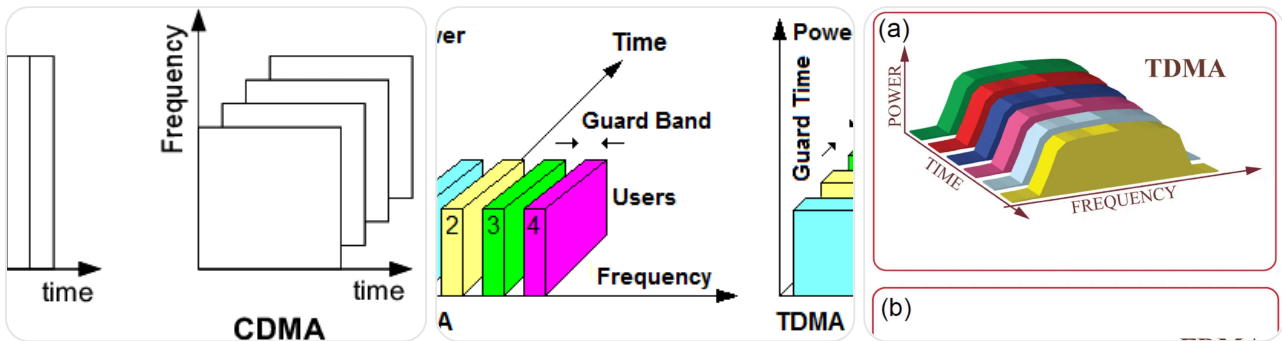
❌ Disadvantages

- Co-channel interference

◆ 3. FDMA, TDMA, CDMA (Detailed Difference)

PPT Pages: FDMA, TDMA, CDMA table UNIT 3 (1)

Diagram



✓ Definitions

- **FDMA** → Users separated by frequency
- **TDMA** → Users separated by time slots
- **CDMA** → Users separated by unique codes

📋 Differences (Write in Exam)

Feature	FDMA	TDMA	CDMA
Principle	Frequency division	Time division	Code division
Resource	Frequency bands	Time slots	Codes
Transmission	Continuous	Burst	Spread spectrum
Synchronization	Not required	Required	Not required
Data Rate	Low	Medium	High
Flexibility	Low	Medium	High
Interference	High	Medium	Low
Guard	Guard bands	Guard time	Both

Feature	FDMA	TDMA	CDMA
Efficiency	Low	Medium	High
Complexity	Simple	Moderate	Complex

🌐 Examples

- FDMA → Radio
- TDMA → GSM
- CDMA → 3G

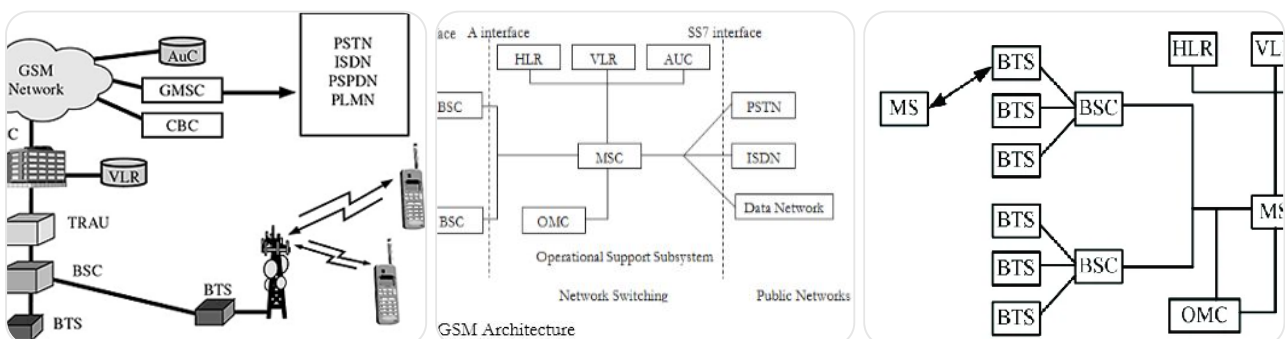
✅ Advantages of CDMA (Extra Point for Exam)

- High capacity
- Better security
- Less interference

◆ 4. GSM Architecture

📄 PPT Pages: GSM Architecture 📁 UNIT 3 (1)

🖼️ Diagram



✅ Definition

GSM (Global System for Mobile Communication) is a **digital cellular standard** used for **mobile communication**.

Components

≡ ChatGPT ▾



-
- Mobile phone + SIM
-

◆ 2. Base Station Subsystem (BSS)

- **BTS (Base Transceiver Station)**
→ Communicates with mobile
 - **BSC (Base Station Controller)**
→ Controls multiple BTS
-

◆ 3. Network Switching Subsystem

- **MSC (Mobile Switching Center)**
→ Call routing
 - **HLR (Home Location Register)**
→ Permanent user data
 - **VLR (Visitor Location Register)**
→ Temporary user data
 - **AuC (Authentication Center)**
→ Security
 - **EIR (Equipment Identity Register)**
→ Device validation
-

Working (Important for Exam)

1. User initiates call
 2. Signal → BTS
 3. BTS → BSC
 4. BSC → MSC
 5. MSC connects to receiver
-

Example

Calling a friend → passes through GSM network components

✓ Advantages

- Secure communication
- Supports roaming
- Efficient

✗ Disadvantages

- Limited data speed

◆ 5. Mobile Radio Propagation

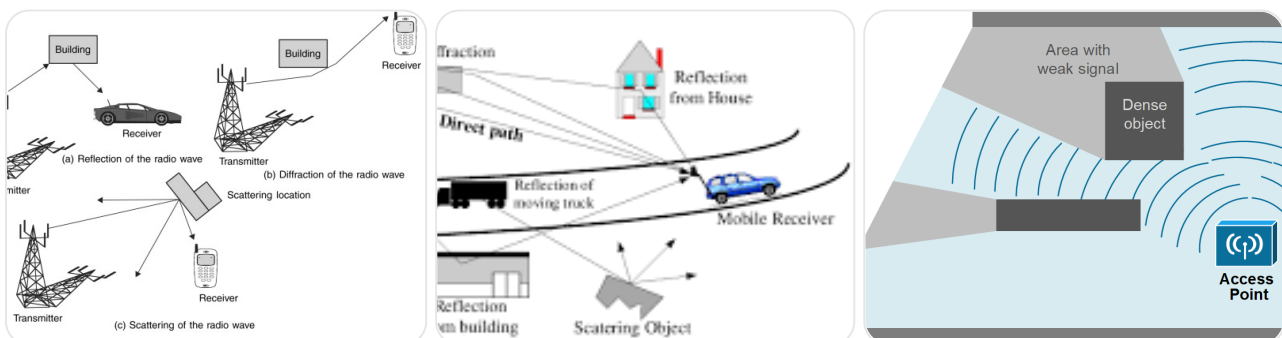
PPT Pages: Mobile Radio Propagation UNIT 3 (1)

✓ Definition

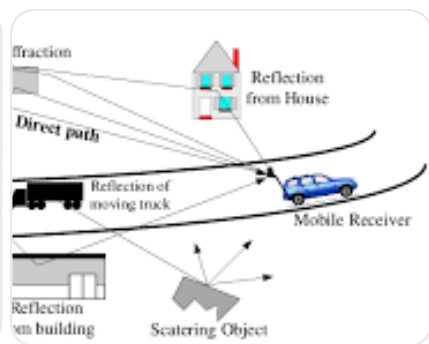
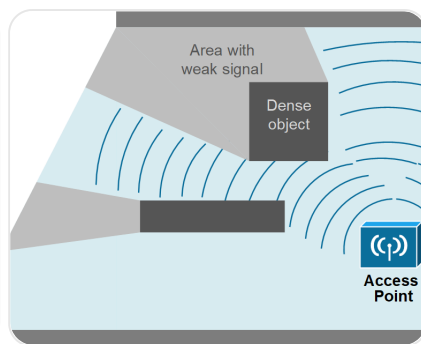
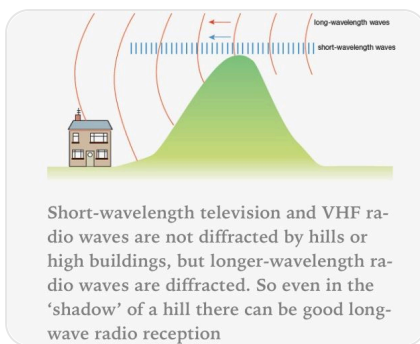
It refers to how radio signals travel from transmitter to receiver in wireless communication.

📊 Diagrams

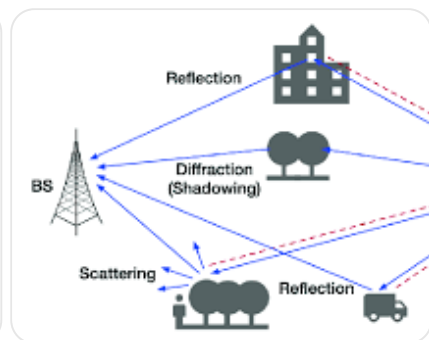
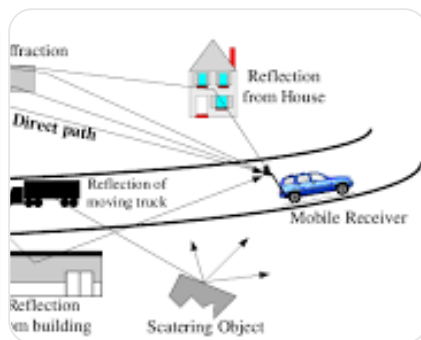
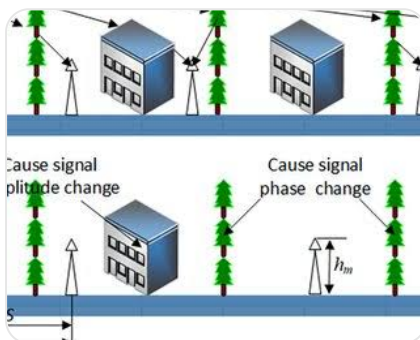
◆ Reflection



◆ Diffraction



◆ Scattering



📌 Types Explained

1. Reflection

- Signal hits surface → bounces back
- Example: Buildings

2. Diffraction

- Signal bends around obstacle
- Example: Hills

3. Scattering

- Signal spreads in many directions
- Example: Trees

📌 Effects

- Multipath propagation
- Signal fading

Example

Mobile signals in city reflect from buildings

Advantages

- Coverage in non-line-of-sight
-

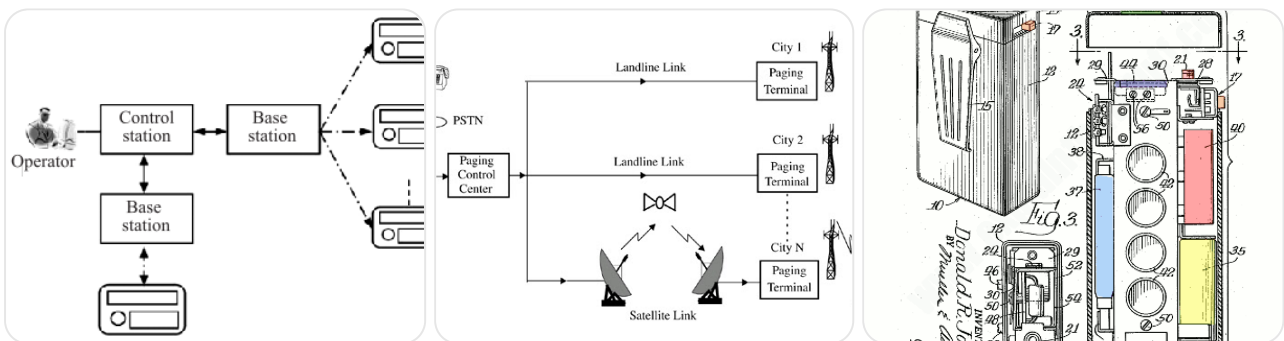
Disadvantages

- Interference
 - Signal distortion
-

6. Paging Systems

 PPT Pages: Paging Systems  UNIT 3 (1)

Diagram



Definition

Paging system is a **simplex (one-way) communication system** used to send short messages.

Working (Step-wise)

1. Sender dials pager number
2. Message goes to control center

3. Broadcast to base stations
4. Pager receives message

📌 Features

- Numeric / alphanumeric messages
- Wide coverage

🌐 Example

Hospital emergency paging

✅ Advantages

- Simple
- Reliable
- Low cost

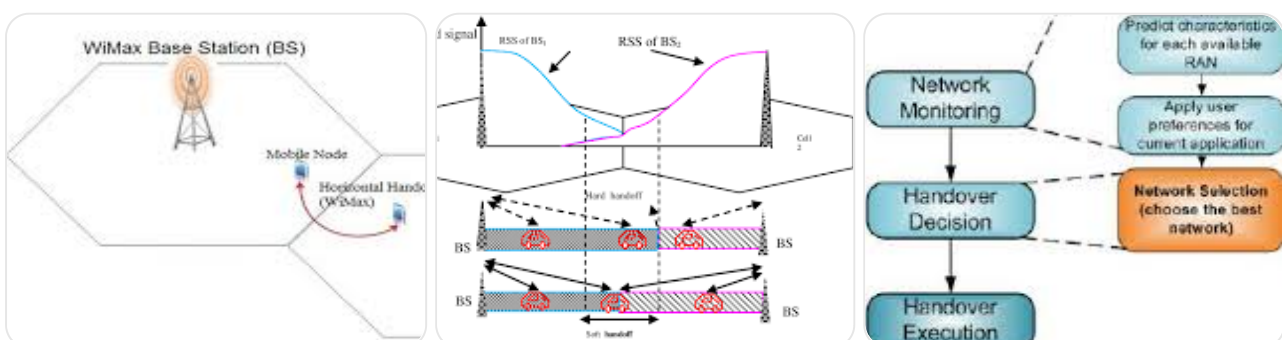
❌ Disadvantages

- One-way communication

◆ 7. Handoff Technology

📄 PPT Pages: Handoff section 📁 UNIT 3 (1)

📊 Diagram



Definition

Handoff is the process of **transferring an ongoing call from one base station to another without interruption.**

Need of Handoff

- User mobility
 - Signal strength decreases
-

Types

Hard Handoff

- Break before make
 - Used in GSM
-

Soft Handoff

- Make before break
 - Used in CDMA
-

Steps

1. Signal becomes weak
 2. Network detects new cell
 3. Connection transferred
-

Example

While travelling in train → call continues smoothly

Advantages

- Continuous communication
 - Better quality
-

✖ Disadvantages

- Complex
 - Call drop possibility
-

🔥 FINAL EXAM STRATEGY

- Draw diagrams → **very important**
 - Write headings + points → **easy marks**
 - GSM + FDMA/TDMA/CDMA → **most asked**
 - Handoff + propagation → **diagram-based questions**
-

If you want next:

- ✓ I can convert this into **perfect handwritten-style notes (PDF)**
- ✓ Or give **important 4/8 mark answers only**
- ✓ Or **last day revision (2 pages)** 🚀

Sources

< 2/2 > 📄 👍 💬 ⬆️ ↺ ...

Is this conversation helpful so far?

